PHY101: Introduction to Physics-1

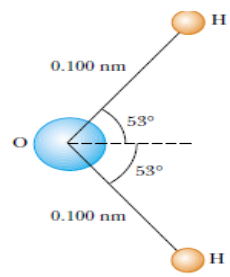
Monsoon, 2018

Tutorial-8

1. A 10.6 kg object oscillates at the end of a vertical spring that has a spring constant of 2.05 x 10^4 N/m. The effect of air resistance is represented by the damping coefficient b= 3.0 N.s/m. (a) Calculate the frequency of the damped oscillation. (b) By what percentage does the amplitude of the oscillation decrease in each cycle? (c) Find the time interval that elapses while the energy of the system drops to 5.0% of its initial value.

2. A mass on the end of a spring (with natural frequency ω) is released from rest at position x_0. The experiment is repeated, but now with the system immersed in a fluid that causes the motion to be overdamped (with damping coefficient γ). Find the ratio of the maximum speed in the former case to that in the latter.

3. A water molecule consists of an oxygen atom with two hydrogen atoms bound to it (see figure). The angle between the two bonds is 106°. If the bonds are 0.1 nm long, where is the center of mass of the molecule?



4. Dr. Srivastava (50 kg) and Dr. Dhar (60 kg) are sitting at the two extremes of a 4 m long boat with mass 40 kg standing still in water. To discuss a research problem, they come to the middle of the boat. How far does the boat move on the water during this process? Neglect the friction of boat with water.